

Homework Unit 4&5 (ET4260) 2026

Semiconductor Sensors & Introduction to Finite Element Modelling

Instructions:

Document your calculations properly with derivations and screenshots when needed.

Please submit your files in word or pdf format. Solutions will be posted one week after submission. Feedback is by self-study.

State here your NAME and STUDENT NUMBER.

Part A: Hall Sensors

Task 1 a):

Considering the material properties given in table 1, which of these intrinsic semiconductors is most suitable to make high-sensitivity Hall sensors? Explain why.

Table 1: Material Properties Si, GaAs, Ge

Property	Si	GaAs	Ge
Atoms (cm^{-3})	5.0×10^{22}	4.42×10^{22}	4.42×10^{22}
Atomic weight	28.09	144.63	72.60
Crystal structure	Diamond	Zincblende	Diamond
Density (g/cm^3)	2.33	5.32	5.33
Lattice constant ($\text{\AA}$)	5.43	5.65	5.65
Melting point ($^{\circ}\text{C}$)	1415	1238	937
Dielectric constant	11.7	13.1	16.0
Bandgap energy (eV)	1.12	1.42	0.66
Electron affinity, χ (V)	4.01	4.07	4.13
Effective density of states in conduction band, N_c (cm^{-3})	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
Effective density of states in valence band, N_v (cm^{-3})	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
Intrinsic carrier concentration (cm^{-3})	1.5×10^{10}	1.8×10^6	2.4×10^{13}
Mobility ($\text{cm}^2/\text{V-s}$)			
Electron, μ_n	1350	8500	3900
Hole, μ_p	480	400	1900

Task 1 b):

Which device parameter links the current-related sensitivity S_I and the voltage-related sensitivity S_V of a Hall sensor to each other? Why must this parameter be considered as well in designing a Hall sensor?

Task 1 c):

When comparing integrated (including circuitry, such as biasing and read-out) Hall devices, which sensitivity is a reasonable figure of merit?

Task 1 d):

Consider the following GaN-Hall devices from paper [1] given in table 2:

Table 2: GaN Hall Sensor Properties

Property	Gen 1	Gen 2	Gen 3	Si
S_I (V/A/T)	68	90	34	300
S_V (mV/V/T)	76	57	34	60
Heterostructure Barrier	AlGaN	AlGaN	InAlN	N/A
Distance between contacts (μm)	200	340	340	100
Shape	Oct	CGC	CGC	Oct
R_{in}, R_{out} (Ω)	870	1560	1030	5350 (0V) to 6500 (2V)
R nonlinearity (%/V)	-10	0.5	0.5	5
Min Offset (<1V)(μT)	500	0.5	2	3
Max Offset (<3V)(μT)	3000	3	4	15
Thermal Noise Floor (nT/ $\sqrt{\text{Hz}}$)	50	89	120	200
Corner Frequency (1V) (kHz)	100 (projected)	0.6	1	7.0

For the Gen 2 devices, the sheet resistance was $R_{\square} = 430 \Omega/\square$.

- Calculate $\left(\frac{L}{W}\right)_{\text{eff}}$ and the geometry factor for the Gen 2 device.
- Given Gen 3 has the same geometry as Gen 2, calculate its' sheet resistance.
- Calculate the sheet carrier density ($n \cdot t$) for Gen 2 and Gen3 in #carriers/cm⁻² as well as the carrier mobility (cm²/Vs). Assume $r = 1$ and $e = 1.602 \times 10^{-19} \text{ C}$.
- Additional task: Compare the sheet resistance and carrier mobility of the silicon device to the values calculated above, assuming $\left(\frac{L}{W}\right)_{\text{eff}} = \sqrt{2}$.

Part B: Temperature Sensors

Task 3 a):

Consider the Wheatstone bridge depicted below. The resistances R_1 and R_4 are moderately n -doped poly-Si and have a negative TCR of $\alpha_{\text{neg}} = -3 \times 10^{-4} \text{ K}^{-1}$ and resistances R_2 and R_3 are highly n -doped poly-Si and thus have a positive TCR of $\alpha_{\text{pos}} = +5 \times 10^{-4} \text{ K}^{-1}$. All resistances are designed to have a value of $R = 1 \text{ k}\Omega$ at 300 K. Find the bias-related sensitivity of the bridge towards changes in temperature.

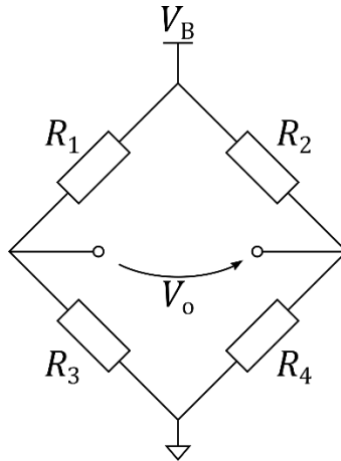


Figure 1: Wheatstone Bridge

Task 3 b):

Compare the statements for output voltage and output rms-noise voltage of a Wheatstone bridge ($R_1 = R_3 = R(1 + \alpha_{\text{neg}} \cdot \Delta T)$, $R_2 = R_4 = R(1 + \alpha_{\text{pos}} \cdot \Delta T)$). How will the SNR generally behave for changing temperature? Assume $\alpha_{\text{neg}} = -\alpha_{\text{pos}}$.

Additional task: plot the exact ($\alpha_{\text{neg}} \neq \alpha_{\text{pos}}$) SNR for $T_0 = 300 \text{ K} \leq T \leq 400 \text{ K}$, assume $ENBW = 1 \text{ kHz}$, $V_B = 5 \text{ V}$ and the R and α -values given in part a). $k_B = 1.381 \times 10^{-23} \text{ J/K}$.

Task 3 c): Additional Task

What measures can be taken so that a measurement with sufficient SNR is possible in the temperature range of interest? What are advantages and disadvantages?

Part C: Photodetectors

Task 2 a):

- Calculate the silicon active device thickness to achieve 90 % and 99 % absorption of light at the wavelengths of 600 nm ($\alpha = 4140 \text{ cm}^{-1}$) and 400 nm ($\alpha = 95200 \text{ cm}^{-1}$).
- Calculate the relative change in quantum efficiency for the thicknesses you calculated for each wave length.
- Calculate the quantum efficiency for each of the wavelengths for 99 % absorption. Assume $\mathcal{R}(400 \text{ nm}) = 0.485$, $\mathcal{R}(600 \text{ nm}) = 0.354$ and $\xi = 1$.

Task 2 b):

Consider a piece of n-doped silicon as a photoconductor at room temperature with the following properties:

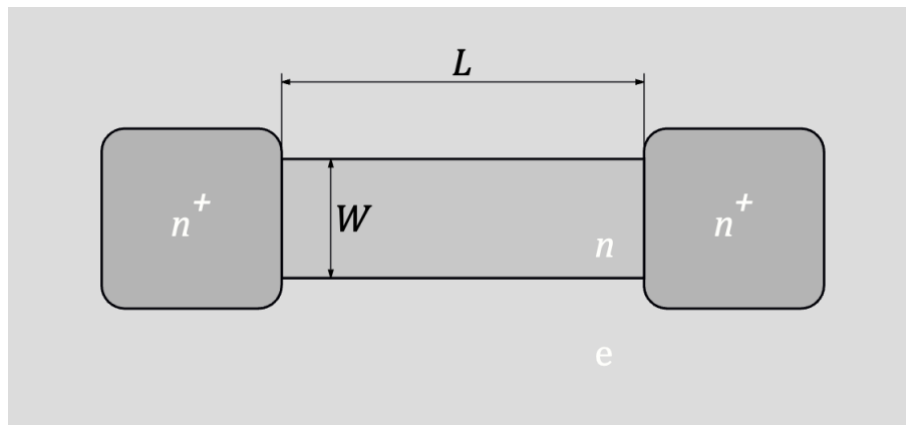


Figure 2: Integrated Photoconductor

Donor concentration of $N_D = 10^{17} \text{ cm}^{-3}$ (assume $n = N_D$) and mobilities $\mu_n = 700 \text{ cm}^2/\text{Vs}$ and $\mu_p = 240 \text{ cm}^2/\text{Vs}$, n-well depth of $t = 1 \text{ }\mu\text{m}$, resistance width of $W = 10 \text{ }\mu\text{m}$ and resistance length of $L = 30 \text{ }\mu\text{m}$.

The photoconductor is driven with a 1.8 V bias (only consider the conductor resistance).

- Calculate the current without optical excitation. The intrinsic carrier concentration is $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, the elementary charge is $e = 1.602 \times 10^{-19} \text{ C}$ and the Boltzmann constant is $k_B = 1.381 \times 10^{-23} \text{ J/K}$.
- Calculate the photocurrent for a photon flux of $\phi = \Phi/(L \cdot W) = 10^{16} \text{ s}^{-1}\text{m}^{-2}$. Assume $\eta = 0.515$ and $\tau = 100 \text{ ns}$.

- iii. Calculate the conductor gain. How much does the gain change in the following situations:
- a) the conductor is twice as long ($2L$)
 - b) the conductor is twice as wide ($2W$)
- iv. How does the photocurrent change for the cases above?

Part D: IS-MOS Capacitance

Task 4 a):

Fill in the gaps for characteristic of a MOS-Capacitance given in the figure below.

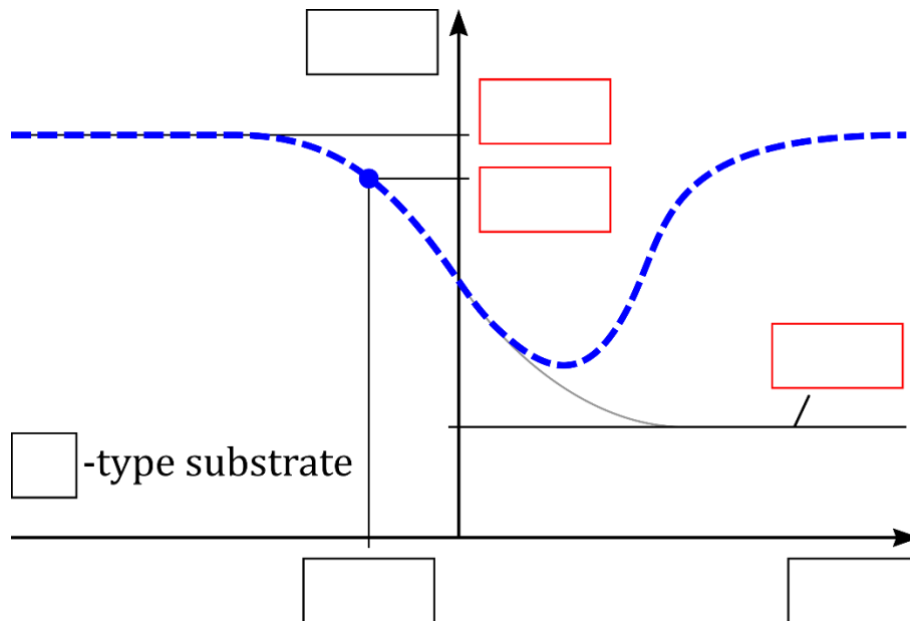


Figure 3: MOS-Cap characteristic

- i. Calculate the three characteristic parameters (red) of the curve at 300 K. Assume the following parameters:

Gate: silicon dioxide SiO_2 , $\epsilon_{\text{SiO}_2, \text{rel}} = 3.9$, $t_{\text{ox}} = 30 \text{ nm}$

Substrate: silicon Si , $\epsilon_{\text{Si}, \text{rel}} = 11.7$, $N_{\text{dopant}} = 3 \times 10^{16} \text{ cm}^{-3}$

The vacuum permittivity is $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$, the intrinsic carrier concentration in Si is $n_i = 1.5 \times 10^{10} \text{ cm}^{-3}$, the elementary charge is $e = 1.602 \times 10^{-19} \text{ C}$ and the Boltzmann constant is $k_B = 1.381 \times 10^{-23} \text{ J/K}$.

- ii. Calculate how these parameters change if instead of SiO_2 the gate oxide is changed to silicon nitride ($\epsilon_{\text{SiN}, \text{rel}} = 7.5$) and the silicon substrates' doping is decreased to $0.5 \times 10^{16} \text{ cm}^{-3}$. Does this change make for a more sensitive sensor? Assume ΔV between C'_{max} and C'_{min} remains the same for both cases.

Task 4 b):

In the image below the blue plot gives an ideal C-V curve for a MOS-Cap with zero charges in the oxide and interface.

Sketch the low-frequency C-V-curve modifications in the plot below for:

- i. fixed (positive) oxide and interface charges (Q_{ox} and Q_{ss})
- ii. semiconductor with a higher work function
- iii. increased pH in a EOS-Cap pH sensor

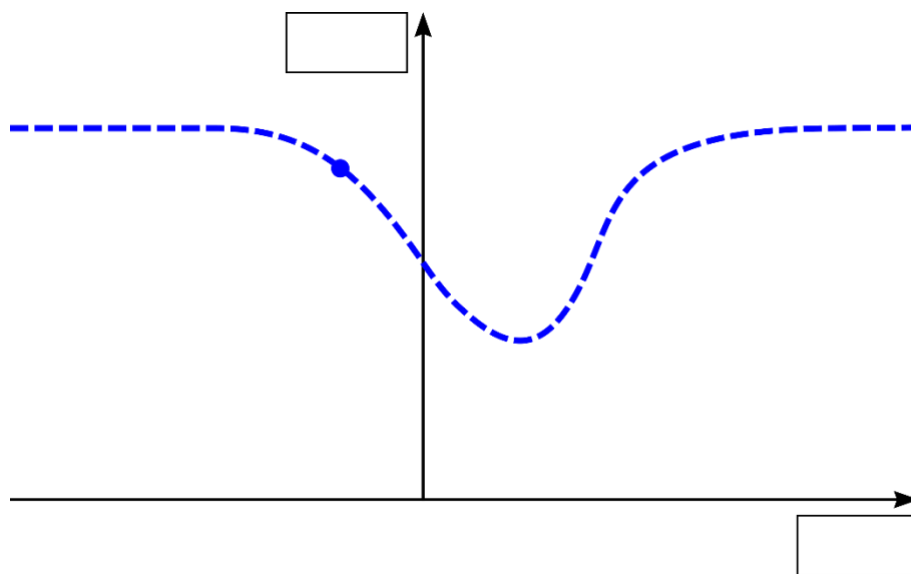


Figure 4: ideal MOS-Cap C-V curve

Task 4 c):

Consider the two EOS-Cap pH-sensors with different oxide material. Calculate the sensor sensitivity ($\Delta C/\Delta pH$) assuming the devices work in a linear regime ($\Delta C/\Delta V = \text{const.}$) with identical slope for both devices. Device A has a SiO_2 gate with a sensitivity parameter of $\alpha_{\text{SiO}_2} = 0.5$ and device B has a Al_2O_3 gate with $\alpha = 0.9$. They work in constant voltage operation.

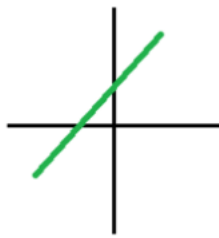
Task 4 d):

How does the sensor response change if you pick another semiconductor material?

Part E: Introduction to FEM Question from Unit 5

Unit 5 Example Question – *this is similar to how we give oral exam questions.*

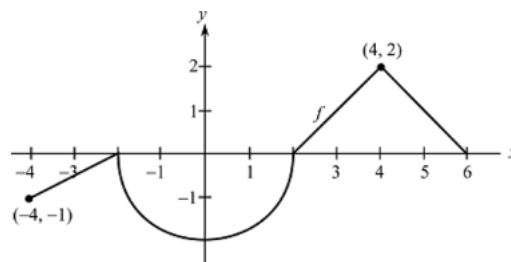
1. Describe the 3 key steps to convert a real physical phenomena to a numerical discretization.
2. What polynomial order of basis function is ideal for FEM, and explain its tradeoff with other functions with a higher or lower order.
3. Propose a qualitative grid spacing to approximate the following curves with linear basis functions (aka hat functions).



(a)



(b)



(c)